Chapter 8 – Introduction to Trigonometry

Question & Answer Sheet

Based strictly on the uploaded NCERT Mathematics Chapter 8 PDF (Reprint 2026–27)

1. What is trigonometry? Explain its meaning and applications mentioned in the chapter.

Answer:

Trigonometry is the branch of mathematics that studies relationships between the sides and angles of a triangle. The word “trigonometry” is derived from the Greek words “tri” (three), “gon” (sides) and “metron” (measure).

The chapter explains that trigonometry can be used to find heights and distances that are difficult to measure directly, such as the height of a monument, the width of a river, or the altitude of a balloon. It is also used in technologically advanced methods in Engineering and Physical Sciences.

The chapter states that early astronomers used trigonometry to find distances of stars and planets from the Earth.

Source: pp. 113–114 (PDF pages 1–2).

2. In a right triangle, define the six trigonometric ratios of an acute angle A.

Answer:

Consider right triangle ABC, right-angled at B. With respect to angle A:

- Hypotenuse = AC
- Side opposite A = BC
- Side adjacent to A = AB

Therefore:

sin A = BC/AC

cos A = AB/AC

tan A = BC/AB

cosec A = AC/BC = 1/sin A

sec A = AC/AB = 1/cos A

cot A = AB/BC = 1/tan A

Thus, cosec, sec and cot are the reciprocals of sin, cos and tan respectively.

Source: pp. 114–115 (PDF pages 2–3).

3. Why do the values of trigonometric ratios of an angle not depend on the size of the right triangle?

Answer:

If two right triangles have the same acute angle, they are similar by the AA similarity criterion. Therefore, their corresponding sides are proportional.

For example, the chapter considers smaller triangles formed inside or along a larger right triangle and shows that ratios such as:

$MP/AP = BC/AC = \sin A$

$AM/AP = AB/AC = \cos A$

$MP/AM = BC/AB = \tan A$

remain unchanged.

Hence, the values of trigonometric ratios depend on the angle, not on the lengths of the sides of the particular right triangle.

Source: pp. 116–117 (PDF pages 4–5).

4. If sin A = 1/3 in a right triangle, find cos A.

Answer:

Given:

$\sin A = 1/3 = BC/AC$.

Let $BC = k$. Then $AC = 3k$.

Using Pythagoras theorem:

$AB^2 = AC^2 - BC^2$

$= (3k)^2 - k^2$

$= 9k^2 - k^2$

$= 8k^2$.

Therefore:

$AB = 2\sqrt{2}\, k$.

Now:

$\cos A = AB/AC$

$= (2\sqrt{2}\, k)/(3k)$

$= 2\sqrt{2}/3$.

Therefore, $\cos A = 2\sqrt{2}/3$.

Source: p. 117 (PDF page 5).

5. Given tan A = 4/3, find all the other trigonometric ratios of A.

Answer:

Given:

$\tan A = 4/3 = BC/AB$.

Let:

$BC = 4k$ and $AB = 3k$.

Using Pythagoras:
$AC^2 = AB^2 + BC^2$
$= (3k)^2 + (4k)^2$
$= 25k^2$.

Hence:
$AC = 5k$.

Therefore:
$\sin A = BC/AC = 4/5$
$\cos A = AB/AC = 3/5$
$\tan A = 4/3$
$\cot A = 3/4$
$\text{cosec } A = 5/4$
$\sec A = 5/3$.

Source: p. 118 (PDF page 6), Example 1.

6. Prove that if two acute angles B and Q satisfy sin B = sin Q, then B = Q.

Answer:

Suppose sin B = sin Q in two right triangles ABC and PQR.

Then:
$\sin B = AC/AB$
$\sin Q = PR/PQ$.

Since $\sin B = \sin Q$:
$AC/AB = PR/PQ$.

Therefore, corresponding sides are proportional. Using Pythagoras theorem, the remaining corresponding sides are also proportional, so:
$AC/PR = AB/PQ = BC/QR$.

Hence the two triangles are similar by the SSS similarity criterion. Therefore, corresponding angles are equal:
$\angle B = \angle Q$.

Thus, B = Q.

Source: pp. 118–119 (PDF pages 6–7), Example 2.

7. In a right triangle ACB, AB = 29 units, BC = 21 units and ∠ABC = θ. Find $\cos^2\theta + \sin^2\theta$ and $\cos^2\theta - \sin^2\theta$.

Answer:

Since the triangle is right-angled at C:
$AC^2 = AB^2 - BC^2$

$= 29^2 - 21^2$
$= (29-21)(29+21)$
$= 8 \times 50$
$= 400.$

Therefore:
$AC = 20.$

For angle θ at B:
$\sin \theta = AC/AB = 20/29$
$\cos \theta = BC/AB = 21/29.$

Therefore:
$\cos^2\theta + \sin^2\theta$
$= (21/29)^2 + (20/29)^2$
$= (441+400)/841$
$= 841/841$
$= 1.$

Also:
$\cos^2\theta - \sin^2\theta$
$= (21^2-20^2)/29^2$
$= (21-20)(21+20)/841$
$= 41/841.$

Answers:
(i) 1
(ii) 41/841.

Source: p. 119 (PDF page 7), Example 3.

8. Find the values of sin 45°, cos 45°, tan 45°, cosec 45°, sec 45° and cot 45°.

Answer:

In a right triangle with one acute angle 45°, the other acute angle is also 45°. Hence the two legs are equal.

Let each leg be a. By Pythagoras:
hypotenuse = $a\sqrt{2}$.

Therefore:
$\sin 45° = a/(a\sqrt{2}) = 1/\sqrt{2}$
$\cos 45° = a/(a\sqrt{2}) = 1/\sqrt{2}$
$\tan 45° = a/a = 1$
$\text{cosec } 45° = \sqrt{2}$
$\sec 45° = \sqrt{2}$

cot 45° = 1.

Source: p. 122 (PDF page 10).

9. Find all six trigonometric ratios for 30° and 60°.

Answer:

From the equilateral-triangle construction in the chapter:

For 30°:
sin 30° = 1/2
cos 30° = √3/2
tan 30° = 1/√3
cosec 30° = 2
sec 30° = 2/√3
cot 30° = √3

For 60°:
sin 60° = √3/2
cos 60° = 1/2
tan 60° = √3
cosec 60° = 2/√3
sec 60° = 2
cot 60° = 1/√3.

Source: p. 123 (PDF page 11) and Table 8.1 on p. 125 (PDF page 13).

10. State the values of the six trigonometric ratios at 0° and 90° and identify the undefined ones.

Answer:

At 0°:
sin 0° = 0
cos 0° = 1
tan 0° = 0
cot 0° = not defined
sec 0° = 1
cosec 0° = not defined

At 90°:
sin 90° = 1
cos 90° = 0
tan 90° = not defined
cot 90° = 0
sec 90° = not defined
cosec 90° = 1.

The undefined values occur because they would require division by zero.

Source: pp. 124–125 (PDF pages 12–13), including Table 8.1.

11. In triangle ABC, right-angled at B, AB = 5 cm and ∠ACB = 30°. Find BC and AC.

Answer:

For angle C = 30°:
AB is opposite to C and BC is adjacent to C.

Using:
tan C = AB/BC.

Therefore:
tan 30° = 5/BC.

Since tan 30° = 1/√3:
1/√3 = 5/BC
BC = 5√3 cm.

Now:
sin 30° = AB/AC
1/2 = 5/AC
AC = 10 cm.

Therefore:
BC = 5√3 cm and AC = 10 cm.

Source: pp. 125–126 (PDF pages 13–14), Example 6.

12. In triangle PQR, right-angled at Q, PQ = 3 cm and PR = 6 cm. Find ∠QPR and ∠PRQ.

Answer:

Since PR is the hypotenuse and PQ is opposite to angle R:
sin R = PQ/PR
= 3/6
= 1/2.

Therefore:
∠PRQ = 30°.

Since the triangle is right-angled at Q, the two acute angles sum to 90°:
∠QPR = 90° − 30° = 60°.

Therefore:
∠QPR = 60°
∠PRQ = 30°.

Source: p. 126 (PDF page 14), Example 7.

13. What is a trigonometric identity? Derive the identity $\sin^2 A + \cos^2 A = 1$.

Answer:

An equation involving trigonometric ratios of an angle is called a trigonometric identity if it is true for all values of the angle(s) for which the expressions are defined.

Consider right triangle ABC, right-angled at B.

By Pythagoras theorem:
$AB^2 + BC^2 = AC^2$.

Divide every term by AC^2:
$AB^2/AC^2 + BC^2/AC^2 = AC^2/AC^2$.

Using:
$AB/AC = \cos A$
$BC/AC = \sin A$

we obtain:
$\cos^2 A + \sin^2 A = 1$.

Therefore:
$\sin^2 A + \cos^2 A = 1$.

Source: p. 128 (PDF page 16), Section 8.4.

14. Derive the identities $1 + \tan^2 A = \sec^2 A$ and $1 + \cot^2 A = \operatorname{cosec}^2 A$.

Answer:

Starting from Pythagoras:
$AB^2 + BC^2 = AC^2$.

For the first identity, divide by AB^2:
$AB^2/AB^2 + BC^2/AB^2 = AC^2/AB^2$.

Therefore:
$1 + \tan^2 A = \sec^2 A$.

For the second identity, divide by BC^2:
$AB^2/BC^2 + BC^2/BC^2 = AC^2/BC^2$.

Therefore:
$\cot^2 A + 1 = \operatorname{cosec}^2 A$.

Hence:
$1 + \tan^2 A = \sec^2 A$

and

$1 + \cot^2 A = \operatorname{cosec}^2 A$.

Source: pp. 128–129 (PDF pages 16–17).

15. If tan A = 1/3, find cot A, sec A, cos A, sin A and cosec A.

Answer:

Given:
tan A = 1/3.

Since cot A = 1/tan A:
cot A = 3.

Using:
$\sec^2 A = 1 + \tan^2 A$,

$\sec^2 A = 1 + (1/3)^2$
$= 1 + 1/9$
$= 10/9$.

However, the supplied chapter's worked example for tan A = 1/3 shows $\sec^2 A = 4/3$, sec A = $2/\sqrt{3}$, cos A = $\sqrt{3}/2$, sin A = 1/2 and cosec A = 2. This corresponds to tan A = $1/\sqrt{3}$, not 1/3.

Therefore, to remain faithful to the chapter's worked example, the intended given value is tan A = $1/\sqrt{3}$, giving:
cot A = $\sqrt{3}$
sec A = $2/\sqrt{3}$
cos A = $\sqrt{3}/2$
sin A = 1/2
cosec A = 2.

Source: p. 129 (PDF page 17), worked example immediately before Example 9. Note: the extracted PDF text renders the numerator/denominator ambiguously; the displayed worked calculation establishes the intended value as tan A = 1/√3.

End of Question & Answer Sheet